Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

[GRADUATION PROJECT PROPOSAL

FarmPulse AI

Submitted by:
Ali Mohab 202300786
Mohamed Hatem 202301610
Mohamed Khaled 202300282
Mazen Ahmed 202300900

Supervised by: Dr. Mohamed Ghalwash

[FALL26]

1. Project Title

FarmPulse AI: AI Multimodal System for Individual Cow Monitoring and Early Health Risk Alerts

2. Team Members

#	Full Name	Student ID	Email	Program
1	Ali Mohab	202300786	s-ali.abozekrey@zewailcity.edu.eg	DSAI
2	Mohamed Khaled	202300282	s-mohamed.elmasry@zewailcity.edu.eg	DSAI
3	Mohamed Hatem	202301610	s-mohamed.shaaban@zewailcity.edu.eg	DSAI
4	Mazen Ahmed	202300900	s-mazen.galal@zewailcity.edu.eg	DSAI

3. Supervisor

Role	Name	Department/Affiliation	Email
Main Supervisor	Dr. Mohamed Ghalwash	Data Science and AI	mghalwash@zewailcity.edu.eg
TA Co-Supervisor	Eng Ahmed Aamer	Data Science and AI	t-ahmed.hamid@zewailcity.edu.eg
Industry Co-Supervisor	Dr. Mohamed Ali Gaber	مدير عام إدارة المجازر تابعة لمديرية الطب البيطري بمحافظة الجيزة	Contact Number: +20 11 25616188 **No email available for now.**

4. Project Classification

Item	Your Project
Research Areas	Artificial Intelligence (multimodal machine learning: computer vision and audio classification), Data Science
Research Focus	Fusing video and audio observations by cow and time to build per-cow records, estimate defined health risks, and generate human-reviewed alerts. Reinforcement learning is explored for decision support once verified outcome data exists.
Applied Field	Agriculture, specifically livestock (cow) monitoring and farm management, especially the cow fattening business.

Item	Your Project
Potential Industrial Partner	None at present. A farm and a veterinarian are used for data collection, veterinarian information, and ground-truth labels. (مزارع العمار، مزارع التؤام)

5. Abstract

Periodic human manual checks on cow farms can miss gradual changes in a cow's behaviour or vocalisation, so early signs of illness or distress often go unnoticed. FarmPulse AI is a proposed multimodal monitoring system that uses camera and microphone data to build a time-stamped record for each cow, flag changes, estimate health risks, and suggest a response for a farmer or veterinarian to review. The system has four modules. A vision module detects behaviour, visible state (such as an open mouth, Rumination process), and possible intruders. A speech module classifies distress sounds and, when reliable, combines them with a visible open mouth to strengthen a stress indication. An analytics module aligns observations by cow and time, stores them, and scores risk, starting with veterinary regulations and moving to a supervised classifier once verified outcomes exist. A reinforcement learning (RL) module evaluates approved response options and proposes one for human review. Intruder events go to a separate immediate security alert. The project also produces a weekly summary and a basic dashboard. The first milestone is a reproducible prototype: a labelled, versioned dataset, baseline vision and audio models evaluated on held-out data, and an end-to-end demonstration of an alert. The risk classifier and RL module will be evaluated only once labelled outcomes and feedback are available. The system converts manual human checks into supervised directioning steps, supports human decisions, and does not diagnose, treat, or claim to predict illness or death.

6. Problem Statement

Farmers and veterinarians till now assess cows through periodic checks, and changes that develop between visits are easy to miss. Examples are altered movement, feeding, or posture, an open mouth, or unusual vocalisation. Individual cows cannot be watched continuously by staff, so there is often no history of how each animal's behaviour has changed over time. Such a history would support earlier attention.

The people affected are farmers, farm staff, and veterinarians who need timely and reliable warning, and the cows whose welfare depends on it. Existing approaches are insufficient because:

- Manual checks are periodic and rely on staff noticing subtle changes.
- Video-only or audio-only monitoring misses information the other signal carries, such as a distress sound with no visible cause.
- Observations are not linked to an individual cow's history, so a single reading cannot be judged against that cow's own baseline.

7. Motivation

- **Technical.** Repeated observations can reveal trends that single checks cannot. Combining video and audio by cow ID and time is a clear multimodal problem with open challenges:

night footage, occlusion, multiple cows in view, background noise, and audio that covers a zone rather than one cow.

- **Economics.** Earlier attention to a possibly ill cow can reduce the cost of missed cases and unnecessary interventions. This is why the proposed reward for the RL stage explicitly weighs missed cases, false alarms, and unneeded interventions.
- **Social and welfare.** Continuous monitoring supports animal welfare, and the system is designed to support human decisions rather than replace farmers or veterinarians.
- **Gap addressed.** The project focuses on a per-cow, time-stamped record that fuses vision and audio, with transparent risk scoring and human-in-the-loop feedback. It is deliberately honest about its limits: risk and RL performance are evaluated only after verified outcomes exist.

8. Objectives

1. **Data foundation (M1).** Collect and label a small, versioned dataset of timestamped video and audio with farm, device, and zone metadata. Document a labelling guide and observation schema, and have a second annotator or domain expert review a subset, reaching agreement of [$\kappa \geq X$].
2. **Perception baseline (M2).** Train baseline vision models for behaviour (CV-M1), visible state (CV-M2), and intruder detection (CV-M3), and a distress-sound classifier (SP-M1). Evaluate them on a held-out test set split by cow and collection period, reporting per-class precision and recall of [$\geq X$] and confusion matrices.
3. **Stress fusion (M2).** Test the rule that sets `stressed=true` when a time-matched distress sound and an open mouth occur together. Report whether it improves on either signal alone.
4. **End-to-end prototype (M3).** Build cow-linked observation records, a transparent rule-based risk baseline, and a demonstrated alert. Measure alert recall, false-alert rate, and lead time on held-out cows or periods, with thresholds chosen with a domain expert.
5. **Reporting and dashboard (M3).** Produce a seven-day weekly report and a basic dashboard showing cow risk, stressed/suspicious counts and reports, plus one traced example from raw clip to recorded outcome.
6. **Decision research (M4).** Define the RL state, actions, and reward. Compare a candidate policy against a simple fixed policy in retrospective or simulated evaluation, with human approval required for any real-world action.

9. Project Scope

Included (In Scope)	Excluded (Out of Scope)
Detecting cow behaviour and visible state from video (movement, feeding, posture, rumination process)	Autonomous treatment, medication, or veterinary diagnosis
Classifying distress or other relevant vocalisations from audio	Claims that illness or death can be predicted accurately before labelled outcomes and a held-out evaluation exist

Included (In Scope)	Excluded (Out of Scope)
Linking observations to a cow and timestamp, and saving a record per cow	Deployment across multiple farms
Rule-based risk scoring, with a supervised classifier only once verified outcomes are available	Fully automated cow identification
A separate immediate security alert for possible intruders	A production-grade dashboard
Weekly summary and basic dashboard (cow risk, stressed/suspicious counts, reports)	Online self-training of the RL policy from unverified alerts
Collecting farmer or veterinarian feedback on alerts and outcomes	Real-world RL actions without human approval
Exploring an RL policy for suggested responses, evaluated offline or in simulation	Treating a risk score or suggested action as a diagnosis

10. Expected Outcomes

Deliverable / Outcome	Description	Target Metric
Labelled, versioned dataset and data contract	Timestamped video and audio with metadata, a labelling guide, a manifest of clips and labels, and a documented observation schema	Inter-annotator agreement of [≥ X] on a reviewed subset; documented dataset version
Baseline vision and audio models	Behaviour, visible-state, intruder and distress-sound models with saved weights, preprocessing settings and results	Per-class precision and recall of [≥ X] on a held-out test set split by cow and period; failure examples reported
Stress fusion rule evaluation	Test of combining a time-matched distress sound with an open mouth	Comparison with each signal alone on the same held-out data
End-to-end prototype	Inference, cow-linked records (cow_profiles, cow_observations, weekly_reports), rule-based risk scoring, alert routing and feedback collection	One traced example from raw clip to alert and recorded outcome; alert recall [≥ X], false-alert rate [≤ Y], lead time [≥ Z]

Deliverable / Outcome	Description	Target Metric
Weekly report and interactive dashboard	A seven-day report built from stored observations, with cow risk and stressed/suspicious counts	Report generated from the previous seven days of data and shown on the dashboard
RL decision-support study	RL formulation, feedback data, offline evaluation and human-reviewed suggestions	Candidate policy compared with a fixed baseline policy in retrospective or simulated evaluation
Documentation and reproducibility	Setup, dependencies, dataset access and evaluation steps	Evaluation reproducible by a third party from the repository

11. Preliminary Literature Review

Summarize 5–10 key related works: what was done, methods used, results, and limitations. Conclude with the research gap your project addresses.

11.1 Summary of Related Work

Ref.	Author(s), Year	Method / Approach	Key Findings	Limitations / Gap
[1]	Seema S. & Ramesh L., 2025	Reinforcement learning with Proximal Policy Optimization (PPO) for adaptive feeding and health management in dairy cattle. It uses multimodal sensor, environmental, and behavioral data, with training in a digital-twin simulation. It was compared with DQN, A2C, Model-Based RL, and Multi-Task Graph Convolutional RL on the MmCows dataset.	PPO gave the best results: 95% milk yield accuracy, 92% health score improvement, 90% feed efficiency, 18% cost reduction, and 0.5 s real-time latency. It outperformed DQN (88%, 80%, 75%, 12%, 0.6 s) and A2C (90%, 85%, 80%, 15%, 0.5 s).	Multimodal data collection needs costly sensors and infrastructure. Training is computationally heavy, farm dynamics are non-stationary, and model interpretability is poor, which limits adoption by non-technical farmers. Generalization across farms is hard, and real-world safety still needs validation.
[2]	Jia Q., Yang J., Han S., Du Z. & Liu J., 2024	CAMLLA-YOLOv8n, an improved YOLOv8n for recognizing seven Holstein cow behaviors (grazing, standing, lying, licking, estrus/mating,	Compared with YOLOv8n, it improved Precision, Recall, mAP@0.5 and mAP@0.5:0.95 by 2.18%, 1.62%, 1.84%	The self-built dataset is small (2,418 images) and comes from a single farm. Classes are highly imbalanced (14,800

Ref.	Author(s), Year	Method / Approach	Key Findings	Limitations / Gap
		fighting, empty bed). It adds Coordinate Attention in the C2f module (C2f-CA), MLLAttention in the neck, an SPPF-GPE module that combines global average and max pooling, and Shape-IoU loss in place of CIoU. It uses Mosaic and other data augmentation.	and 1.77%, reaching 94.46%, 94.16%, 94.04% and 92.97%. It also beat YOLOv3-tiny, YOLOv5n/s, YOLOv7-tiny and YOLOv8s. Inference took 35.5 ms per image, and it ran in real time on Jetson AGX Xavier and Atlas 200 DK edge devices.	lying boxes versus 16 mating and 70 licking). Parameters and FLOPs rose slightly over YOLOv8n (3.254 M vs 3.012 M; 8.6 G vs 8.2 G). The paper names model compression (pruning and quantization) as future work for edge deployment.
[3]	Sim H.-S. & Cho H.-C., 2025	Enhanced multi-cattle tracking using tracking-by-detection. YOLOv9-t is the detector, and five bounding-box losses (GIoU, DIoU, CIoU, SIoU, WIoU) were compared. DeepSORT and StrongSORT are the trackers. The Kalman filter state vector uses bounding-box width instead of aspect ratio, and the Re-ID distance metric is Pearson distance instead of cosine distance.	DIoU gave the best detection (mAP 95.3%). With DeepSORT, the combined “width + Pearson” setting raised HOTA by 4.1% to 78.64%, MOTA by 1.08% to 90.29%, and IDF1 by 5.12% to 91.41%, and ID switches fell from 18 to 5. StrongSORT started higher (HOTA 78.53%) and reached 79.56% with the changes.]	The data come from only four cattle (two cows and two calves) in one controlled research pen with a single camera. The tracking test used just eight 10-second clips. Occlusions were reduced but not eliminated. The authors suggest multi-camera setups and more diverse, larger farm data as future work.
[4]	Sim & Cho, 2025	Tracking-by-detection using a YOLOv9-t detector with DeepSORT and StrongSORT trackers. They compared 5 IoU losses (GIoU, DIoU, CIoU, SIoU, WIoU), replaced aspect ratio with box width in the Kalman filter state, and replaced cosine distance with Pearson distance for Re-ID.	DIoU gave the best detection (mAP 95.3%). With DeepSORT, width + Pearson reached HOTA 78.64%, MOTA 90.29% and IDF1 91.41%, and ID switches dropped from 18 to 5.	Only 4 cattle (2 cows, 2 calves) in one controlled research pen. Single camera, so occlusions were not fully solved. Evaluated on only 8 ten-second clips. Behavior was not recognized.
[5]	Li et al., 2024	Built CBVD-5, a video dataset of 107 cows from 7 cameras: 687 clips and 206,100 frames in AVA format. It covers 5 behaviors (standing, lying, foraging, drinking, rumination). A SlowFast	Test mAP was about 71–74%. AP was high for standing (98.6%), lying (96.7%) and foraging (96.9%), but low for drinking (34.8%) and rumination (41.0%).	Severe class imbalance (drinking is 5.5% of samples). Behaviors are recognized per frame and region, not linked to individual cow IDs; ID tracking is left as

Ref.	Author(s), Year	Method / Approach	Key Findings	Limitations / Gap
		(ResNet-50) model from MMAction2 served as the baseline.		future work. Single farm.
[6]	Yu et al., 2025	Built MultiCamCows2024 from 3 ceiling cameras over 7 days: 101,329 images of 90 Holstein-Friesians. A ResNet-152 rotated-box detector produced tracklets. ResNet-18 Re-ID was trained in a supervised setup (cross-entropy) and a self-supervised setup (NT-Xent + KNN).	Fusing the cameras gave 96.06% accuracy (supervised) and 96.36% (self-supervised), with no manual ID labelling needed.	Closed-set only, so new cows cannot be handled. Tracklets still need human checking. Cows pass once along a narrow milk-race. Accuracy falls in crowded or stationary areas (Camera 3: 56.69%). No behavior analysis.
[7]	Zhang Y., Zhang Y., Jiang H., Du H., Xue A. & Shen W., 2024	A multimodal cow-behaviour system combining barn video and neck-collar IMU data. EfficientNetV2-S extracted video features, BiLSTM processed sensor sequences, and a Transformer fused both modalities to classify drinking, feeding, lying, standing, and walking.	On five Holstein cows and 228 labelled clips, multimodal fusion achieved 98.80% accuracy, 97.15% precision, and 96.93% recall. It outperformed the video-only and sensor-only models, showing that complementary signals can improve behaviour recognition.	The experiment used only five cows in a controlled setting and relied on a collar that supplied identity-related information. It did not use audio, did not address multi-cow occlusion or long-term re-identification, and did not validate behaviour changes against veterinary outcomes.
[8]	Li G., Xiong Y., Du Q., Shi Z. & Gates R. S., 2021	Automatic recognition of ingestive jaw activities from head-mounted microphone recordings. The study extracted MFCC acoustic features and compared 1D-CNN, 2D-CNN, and LSTM models for classifying bites, chews, and chew-bites.	The best LSTM model achieved an overall F1-score of 0.881; performance was strongest for chews (F1 = 0.915). The study shows that audio can provide useful evidence of feeding-related jaw activity.	Only three cows were recorded, using head-mounted microphones in controlled grazing conditions. Performance changed across forage species and height. The approach was not tested with noisy barn microphones, did not associate sound with a visible individual cow, and did not fuse audio

Ref.	Author(s), Year	Method / Approach	Key Findings	Limitations / Gap
				with video or health outcomes.
[9]	Jobarteh B., Mincu-Iorga M., Gavojdian D. & Neethirajan S., 2025	The study fused conventional acoustic features, such as frequency, duration, and amplitude, with Whisper-derived non-linguistic vocalisation motifs. SVM, Random Forest, and RNN models classified high-frequency and low-frequency cattle vocalisations recorded during short social isolation.	The SVM achieved 98.35% accuracy in separating high-frequency and low-frequency calls. Combining acoustic and learned vocalisation features improved the analysis of cattle vocal signals.	The labels represent vocalisation categories in one controlled isolation context, not verified illness or distress diagnoses. Data came from 20 cows on one farm and contained class imbalance. The work did not connect calls to video observations, individual cow identity, or veterinary-confirmed health outcomes
[10]	Wu D., Wang H., Han S., Song J., Shang L., Zhang D. & Song H., 2021	A VGG16-CNN and BiLSTM model was used to recognise drinking, ruminating, walking, standing, and lying from video. The study evaluated behaviour recognition under day/night, weather, and wind variations.	The model achieved an average accuracy of 97.6% on the reported test videos. Results suggest that video-based behaviour recognition can remain effective under several environmental conditions.	The camera was deliberately configured so that only one cow appeared in each video; therefore, the study does not solve multi-cow detection, identity continuity, overlap, or re-identification. It also used video only, had no audio fusion, and did not link detected behaviour to verified health outcomes.

11.2 Research Gap

Existing studies demonstrate important individual components of intelligent cow monitoring, but they do not yet provide a reliable end-to-end process for maintaining a per-cow, multimodal history in realistic farm conditions. Vision research can recognise behaviours such as standing, lying, feeding, walking, drinking, and rumination; however, high performance is often reported when a single cow is

deliberately framed or when behaviour is recognised at image or region level rather than being linked continuously to the same individual. Multi-cow tracking and re-identification research improves identity continuity, but published evaluations are still limited by small herds, short clips, single-camera views, controlled pens, crowded scenes, or restricted cow movement. Therefore, occlusion, similar-looking cows, ID switches, and re-identification after a cow leaves and re-enters the camera view remain practical challenges.

Audio studies show that jaw sounds and vocalisation patterns can be classified, but their conditions are restrictive. Some methods require microphones attached to the cow, while others use controlled social-isolation recordings. A barn microphone usually covers an area containing several cows and background machinery, so a detected sound cannot automatically be assigned to one animal. In addition, vocalisation categories or unusual sounds should not be treated as a veterinary diagnosis without verified clinical labels. Current studies rarely align an audio event, a visible behaviour, a stable cow identity, and a veterinary-confirmed outcome in one longitudinal record.

FarmPulse AI addresses this gap through a cautious, human-in-the-loop prototype. The project will synchronise camera and microphone observations by time and farm zone; detect and track cows in video; and create a time-stamped record only when the cow identity is sufficiently reliable. When an audio event cannot be reliably linked to one cow, it will remain a zone-level event rather than being assigned incorrectly. The system will evaluate whether a time-matched vocal event and visible cue, such as an open mouth or abnormal behaviour, is more useful than either signal alone.

The project will not claim autonomous diagnosis, treatment, or guaranteed early disease prediction. It will first use transparent risk rules and will train a supervised risk model only after farmer or veterinarian feedback provides verified outcomes. Evaluation will use held-out data separated by cow and time period, and will report per-class precision, recall, ID-switch and false-alert cases, rather than relying only on overall accuracy. Thus, FarmPulse AI contributes a reproducible multimodal monitoring workflow that connects perception, identity-aware records, cautious risk alerts, and human review.

12. Timeline

Initial phases includes our data collection journey, data labelling, and veterinary to let our team ready for a clear start without any missleading veterinary information and regulations. Also consider that we are collecting real live streams manually.

#	Phase	Key Activities	Start	End	Milestone
1	Project Proposal	Identify our scope problem, and clarify our main goals	1 OCT 2026	22 OCT 2026	Proposal
2	Literature Review & Market Visibility	Review research on cow behaviour recognition, rumination and vocalisation analysis, cattle tracking/re-identification, and multimodal AI fusion. Review existing farm-monitoring solutions and identify market needs through discussions with farmers and veterinarians, including practical camera/audio setup, alert requirements, dashboard needs,	22 OCT 2026	31 Oct 2026	M1

#	Phase	Key Activities	Start	End	Milestone
		cost, and data-collection constraints. Finalise the research gap, system requirements, and feasible project scope.			
3	Data Collection & Labeling	**Veterinary data collection:** • Collect veterinary information. • Learn the visionary and vocal indications for illness. • Cover the business needs, shortages, and lacks **Technical data collection:** • Live day recording. • Cutting videos. • Labeling and annotating our own collected data.	1 NOV 2026	7 DEC 2026	M1
4	Naive Models creation	• Models selection. • Features extraction. • Train and test each model individually.	1 DEC 2026	1 JAN 2027	M2 & M3
5	Naive Models Integration	After evaluating some models, start integrating them and validate	10 DEC 2026	10 JAN 2027	M3 & M4

References

Use a consistent citation style (IEEE recommended), numbered in order of appearance.

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